Octal D-Type Flip-Flop with 3-State Outputs

MC74AC374, MC74ACT374

The MC74AC374/74ACT374 is a high−speed, low−power octal D−type flip−flop featuring separate D−type inputs for each flip−flop and 3−state outputs for bus−oriented applications. A buffered Clock (CP) and Output Enable ($\overline{OE}$) are common to all flip−flops.

Features

- Buffered Positive Edge−Triggered Clock
- 3−State Outputs for Bus−Oriented Applications
- Outputs Source/Sink 24 mA
- See MC74AC273 for Reset Version
- See MC74AC377 for Clock Enable Version
- See MC74AC373 for Transparent Latch Version
- See MC74AC574 for Broadside Pinout Version
- See MC74AC564 for Broadside Pinout Version with Inverted Outputs
- ′ACT374 Has TTL Compatible Inputs
- These are Pb−Free Devices

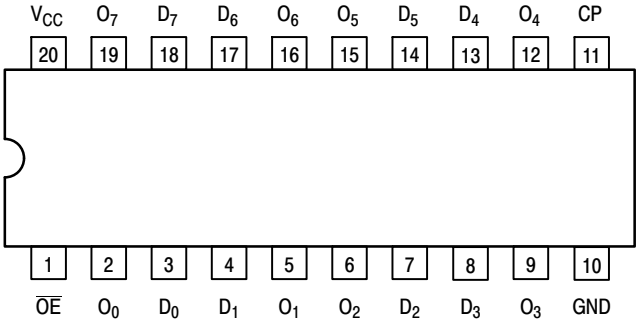


Figure 1. Pinout: 20 Lead Packages Conductors

(Top View)

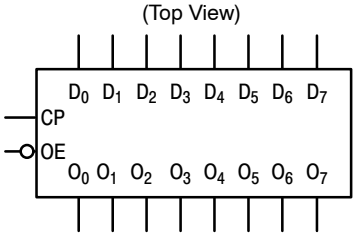
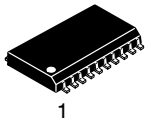


Figure 2. Logic Symbol

PIN ASSIGNMENT

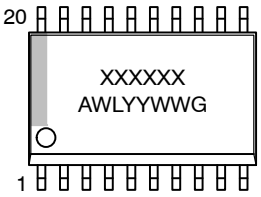
PIN	FUNCTION
D_0−D_7	Data Inputs
CP	Clock Pulse Input
$\overline{OE}$	3−State Output Enable Input
O_0−O_7	3−State Outputs

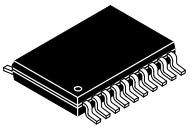
MARKING DIAGRAMS



1

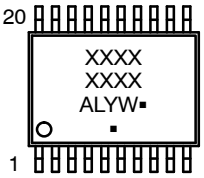
SOIC−20W
DW SUFFIX
CASE 751D





1

TSSOP−20
DT SUFFIX
CASE 948E



XXXXXX = Specific Device Code
A = Assembly Location
WL, L = Wafer Lot
YY, Y = Year
WW, W = Work Week
G or ▪ = Pb−Free Package

(Note: Microdot may be in either location)

DEVICE MARKING INFORMATION

See general marking information in the device marking section on page 6 of this data sheet.

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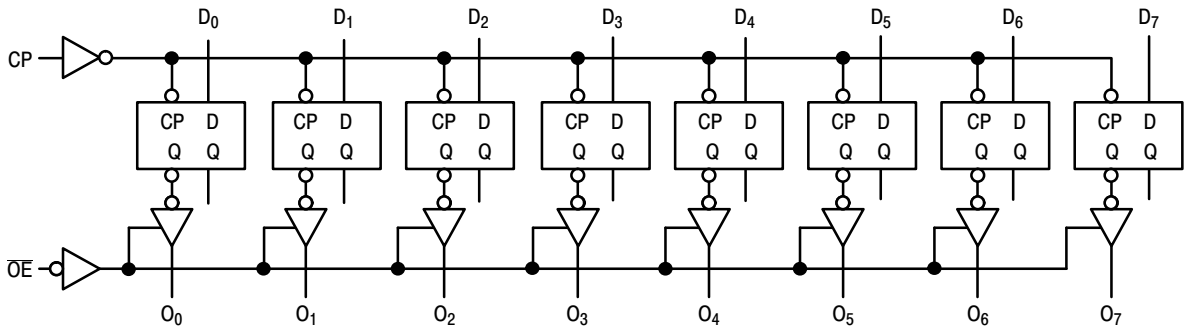
TRUTH TABLE

Inputs			Outputs
D_n	CP	$\overline{OE}$	O_n
H	⌐¬	L	H
L	⌐¬	L	L
X	X	H	Z

H = HIGH Voltage Level
L = LOW Voltage Level
X = Immaterial
Z = High Impedance
⌐¬ = LOW-to-HIGH Transition

FUNCTIONAL DESCRIPTION

The MC74AC374/74ACT374 consists of eight edge−triggered flip−flops with individual D−type inputs and 3−state true outputs. The buffered clock and buffered Output Enable are common to all flip−flops. The eight flip−flops will store the state of their individual D inputs that meet the setup and hold time requirements on the LOW−to−HIGH Clock (CP) transition. With the Output Enable ($\overline{OE}$) LOW, the contents of the eight flip−flops are available at the outputs. When the $\overline{OE}$ is HIGH, the outputs go to the high impedance state. Operation of the $\overline{OE}$ input does not affect the state of the flip−flops.



NOTE: That this diagram is provided only for the understanding of logic operations and should not be used to estimate propagation delays.

Figure 3. Logic Diagram

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MAXIMUM RATINGS

Symbol	Parameter		Value	Unit
V_{CC}	DC Supply Voltage (Referenced to GND)		−0.5 to +6.5	V
V_{IN}	DC Input Voltage (Referenced to GND)		−0.5 to V_{CC} +0.5	V
V_{OUT}	DC Output Voltage (Referenced to GND) (Note 1)		−0.5 to V_{CC} +0.5	V
I_{IK}	DC Input Diode Current		±20	mA
I_{OK}	DC Output Diode Current		±50	mA
I_{OUT}	DC Output Sink/Source Current		±50	mA
I_{CC}	DC Supply Current, per Output Pin		±50	mA
I_{GND}	DC Ground Current, per Output Pin		±100	mA
T_{STG}	Storage Temperature Range		−65 to +150	°C
T_L	Lead temperature, 1 mm from Case for 10 Seconds		260	°C
T_J	Junction Temperature Under Bias		140	°C
θ_{JA}	Thermal Resistance (Note 2)	SOIC TSSOP	96 150	°C/W
MSL	Moisture Sensitivity		Level 1	
F_R	Flammability Rating	Oxygen Index: 30% − 35%	UL 94 V−0 @ 0.125 in	
V_{ESD}	ESD Withstand Voltage	Human Body Model (Note 3) Charged Device Model (Note 4)	> 2000 > 1000	V
$I_{Latchup}$	Latchup Performance	Above V_{CC} and Below GND at 85°C (Note 5)	±100	mA

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

1. I_{OUT} absolute maximum rating must be observed.
2. The package thermal impedance is calculated in accordance with JESD 51−7.
3. Tested to EIA/JESD22−A114−A.
4. Tested to JESD22−C101−A.
5. Tested to EIA/JESD78.

RECOMMENDED OPERATING CONDITIONS

Symbol	Parameter		Min	Typ	Max	Unit
V_{CC}	Supply Voltage	′AC	2.0	5.0	6.0	V
		′ACT	4.5	5.0	5.5	
V_{IN}, V_{OUT}	DC Input Voltage, Output Voltage (Ref. to GND)		0		V_{CC}	V
t_r, t_f	Input Rise and Fall Time (Note 1) ′AC Devices except Schmitt Inputs	V_{CC} @ 3.0 V	−	150	−	ns/V
		V_{CC} @ 4.5 V	−	40	−	
		V_{CC} @ 5.5 V	−	25	−	
t_r, t_f	Input Rise and Fall Time (Note 2) ′ACT Devices except Schmitt Inputs	V_{CC} @ 4.5 V	−	10	−	ns/V
		V_{CC} @ 5.5 V	−	8.0	−	
T_A	Operating Ambient Temperature Range		−40	25	85	°C
I_{OH}	Output Current − High		−	−	−24	mA
I_{OL}	Output Current − Low		−	−	24	mA

Functional operation above the stresses listed in the Recommended Operating Ranges is not implied. Extended exposure to stresses beyond the Recommended Operating Ranges limits may affect device reliability.

1. V_{IN} from 30% to 70% V_{CC}; see individual Data Sheets for devices that differ from the typical input rise and fall times.
2. V_{IN} from 0.8 V to 2.0 V; see individual Data Sheets for devices that differ from the typical input rise and fall times.

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DC CHARACTERISTICS

Symbol	Parameter	V_{CC} (V)	74AC T_A = +25°C	74AC T_A = −40°C to +85°C		Unit	Conditions
			Typ	Guaranteed Limits			
V_{IH}	Minimum High Level Input Voltage	3.0 4.5 5.5	1.5 2.25 2.75	2.1 3.15 3.85	2.1 3.15 3.85	V	V_{OUT} = 0.1 V or V_{CC} − 0.1 V
V_{IL}	Maximum Low Level Input Voltage	3.0 4.5 5.5	1.5 2.25 2.75	0.9 1.35 1.65	0.9 1.35 1.65	V	V_{OUT} = 0.1 V or V_{CC} − 0.1 V
V_{OH}	Minimum High Level Output Voltage	3.0 4.5 5.5	2.99 4.49 5.49	2.9 4.4 5.4	2.9 4.4 5.4	V	I_{OUT} = −50 μA
		3.0 4.5 5.5	− − −	2.56 3.86 4.86	2.46 3.76 4.76	V	*V_{IN} = V_{IL} or V_{IH} −12 mA I_{OH} −24 mA −24 mA
V_{OL}	Maximum Low Level Output Voltage	3.0 4.5 5.5	0.002 0.001 0.001	0.1 0.1 0.1	0.1 0.1 0.1	V	I_{OUT} = 50 μA
		3.0 4.5 5.5	− − −	0.36 0.36 0.36	0.44 0.44 0.44	V	*V_{IN} = V_{IL} or V_{IH} 12 mA I_{OL} 24 mA 24 mA
I_{IN}	Maximum Input Leakage Current	5.5	−	±0.1	±1.0	μA	V_I = V_{CC}, GND
I_{OZ}	Maximum 3-State Current	5.5	−	±0.5	±5.0	μA	V_I (OE) = V_{IL}, V_{IH} V_I = V_{CC}, GND V_O = V_{CC}, GND
I_{OLD}	†Minimum Dynamic Output Current	5.5	−	−	75	mA	V_{OLD} = 1.65 V Max
I_{OHD}		5.5	−	−	−75	mA	V_{OHD} = 3.85 V Min
I_{CC}	Maximum Quiescent Supply Current	5.5	−	8.0	80	μA	V_{IN} = V_{CC} or GND

*All outputs loaded; thresholds on input associated with output under test.
†Maximum test duration 2.0 ms, one output loaded at a time.
NOTE: I_{IN} and I_{CC} @ 3.0 V are guaranteed to be less than or equal to the respective limit @ 5.5 V V_{CC}.

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AC CHARACTERISTICS (For Figures and Waveforms – See AND8277/D at www.onsemi.com)

Symbol	Parameter	V_{CC}* (V)	74AC T_A = +25°C C_L = 50 pF			74AC T_A = –40°C to +85°C C_L = 50 pF		Unit	Fig. No.
			Min	Typ	Max	Min	Max		
f_{max}	Maximum Clock Frequency	3.3 5.0	60 100	110 155	– –	60 100	– –	MHz	3–3
t_{PLH}	Propagation Delay CP to O_n	3.3 5.0	3.0 2.5	11 8.0	13.5 9.5	1.5 1.5	15.5 10.5	ns	3–6
t_{PHL}	Propagation Delay CP to O_n	3.3 5.0	2.5 2.0	10 7.0	12.5 9.0	2.0 1.5	14 10	ns	3–6
t_{PZH}	Output Enable Time	3.3 5.0	3.0 2.0	9.5 7.0	11.5 8.5	1.5 1.0	13 9.5	ns	3–7
t_{PZL}	Output Enable Time	3.3 5.0	2.5 2.0	9.0 6.5	11.5 8.5	1.5 1.0	13 9.5	ns	3–8
t_{PHZ}	Output Disable Time	3.3 5.0	3.0 2.0	10.5 8.0	12.5 11	2.0 2.0	14.5 12.5	ns	3–7
t_{PLZ}	Output Disable Time	3.3 5.0	2.0 1.5	8.0 6.5	11.5 8.5	1.0 1.0	12.5 10	ns	3–8

*Voltage Range 3.3 V is 3.3 V ±0.3 V.
Voltage Range 5.0 V is 5.0 V ±0.5 V.

AC OPERATING REQUIREMENTS

Symbol	Parameter	V_{CC}* (V)	74AC T_A = +25°C C_L = 50 pF		74AC T_A = –40°C to +85°C C_L = 50 pF	Unit	Fig. No.
			Typ	Guaranteed Minimum			
t_s	Setup Time, HIGH or LOW D_n to CP	3.3 5.0	2.0 1.0	5.5 4.0	6.0 4.5	ns	3–9
t_h	Hold Time, HIGH or LOW D_n to CP	3.3 5.0	–1.0 0	1.0 1.5	1.0 1.5	ns	3–9
t_w	CP Pulse Width HIGH or LOW	3.3 5.0	4.0 2.5	5.5 4.0	6.0 4.5	ns	3–6

*Voltage Range 3.3 V is 3.3 V ±0.3 V.
Voltage Range 5.0 V is 5.0 V ±0.5 V.

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DC CHARACTERISTICS

Symbol	Parameter	V_{CC} (V)	74ACT T_A = +25°C		74ACT T_A = −40°C to +85°C	Unit	Conditions
			Typ	Guaranteed Limits			
V_{IH}	Minimum High Level Input Voltage	4.5 5.5	1.5 1.5	2.0 2.0	2.0 2.0	V	V_{OUT} = 0.1 V or V_{CC} − 0.1 V
V_{IL}	Maximum Low Level Input Voltage	4.5 5.5	1.5 1.5	0.8 0.8	0.8 0.8	V	V_{OUT} = 0.1 V or V_{CC} − 0.1 V
V_{OH}	Minimum High Level Output Voltage	4.5 5.5	4.49 5.49	4.4 5.4	4.4 5.4	V	I_{OUT} = −50 μA
		4.5 5.5	– –	3.86 4.86	3.76 4.76	V	*V_{IN} = V_{IL} or V_{IH} I_{OH} −24 mA −24 mA
V_{OL}	Maximum Low Level Output Voltage	4.5 5.5	0.001 0.001	0.1 0.1	0.1 0.1	V	I_{OUT} = 50 μA
		4.5 5.5	– –	0.36 0.36	0.44 0.44	V	*V_{IN} = V_{IL} or V_{IH} I_{OL} 24 mA 24 mA
I_{IN}	Maximum Input Leakage Current	5.5	–	±0.1	±1.0	μA	V_I = V_{CC}, GND
ΔI_{CCT}	Additional Max. I_{CC}/Input	5.5	0.6	–	1.5	mA	V_I = V_{CC} − 2.1 V
I_{OZ}	Maximum 3-State Current	5.5	–	±0.5	±5.0	μA	V_I (OE) = V_{IL}, V_{IH} V_I = V_{CC}, GND V_O = V_{CC}, GND
I_{OLD}	†Minimum Dynamic Output Current	5.5	–	–	75	mA	V_{OLD} = 1.65 V Max
I_{OHD}		5.5	–	–	−75	mA	V_{OHD} = 3.85 V Min
I_{CC}	Maximum Quiescent Supply Current	5.5	–	8.0	80	μA	V_{IN} = V_{CC} or GND

*All outputs loaded; thresholds on input associated with output under test.
†Maximum test duration 2.0 ms, one output loaded at a time.

AC CHARACTERISTICS (For Figures and Waveforms − See AND8277/D at www.onsemi.com)

Symbol	Parameter	V_{CC}* (V)	74ACT T_A = +25°C, C_L = 50 pF			74ACT T_A = −40°C to +85°C, C_L = 50 pF		Unit	Fig. No.
			Min	Typ	Max	Min	Max		
f_{max}	Maximum Clock Frequency	5.0	100	160	–	90	–	MHz	3–3
t_{PLH}	Propagation Delay CP to O_n	5.0	2.0	8.5	10	2.0	11.5	ns	3–6
t_{PHL}	Propagation Delay CP to O_n	5.0	2.0	8.0	9.5	1.5	11	ns	3–6
t_{PZH}	Output Enable Time	5.0	2.0	8.0	9.5	1.5	10.5	ns	3–7
t_{PZL}	Output Enable Time	5.0	1.5	8.0	9.0	1.5	10.5	ns	3–8
t_{PHZ}	Output Disable Time	5.0	1.5	8.5	11.5	1.0	12.5	ns	3–7
t_{PLZ}	Output Disable Time	5.0	1.5	7.0	8.5	1.0	10	ns	3–8

*Voltage Range 5.0 V is 5.0 V ±0.5 V.

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AC OPERATING REQUIREMENTS (For Figures and Waveforms – See AND8277/D at www.onsemi.com)

Symbol	Parameter	V_{CC}* (V)	74ACT T_A = +25°C C_L = 50 pF		74ACT T_A = −40°C to +85°C C_L = 50 pF	Unit	Fig. No.
			Typ	Guaranteed Minimum			
t_s	Setup Time, HIGH or LOW D_n to CP	5.0	1.0	5.0	5.5	ns	3–9
t_h	Hold Time, HIGH or LOW D_n to CP	5.0	0	1.5	1.5	ns	3–9
t_w	CP Pulse Width HIGH or LOW	5.0	2.5	5.0	5.0	ns	3–6

*Voltage Range 5.0 V is 5.0 V ±0.5 V.

CAPACITANCE

Symbol	Parameter	Value Typ	Unit	Test Conditions
C_{IN}	Input Capacitance	4.5	pF	V_{CC} = 5.0 V
C_{PD}	Power Dissipation Capacitance	80	pF	V_{CC} = 5.0 V

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ORDERING INFORMATION

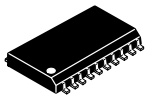
Device	Markiing	Package	Shipping†
MC74AC374DWG	AC374	SOIC−20	38 Units / Rail
MC74AC374DWR2G	AC374	SOIC−20	1000 / Tape & Reel
MC74ACT374DWG	ACT374	SOIC−20	38 Units / Rail
MC74ACT374DWR2G	ACT374	SOIC−20	1000 / Tape & Reel
MC74AC374DTR2G	AC 374	TSSOP−20	2500 / Tape & Reel
MC74ACT374DTR2G	ACT 374	TSSOP−20	2500 / Tape & Reel

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

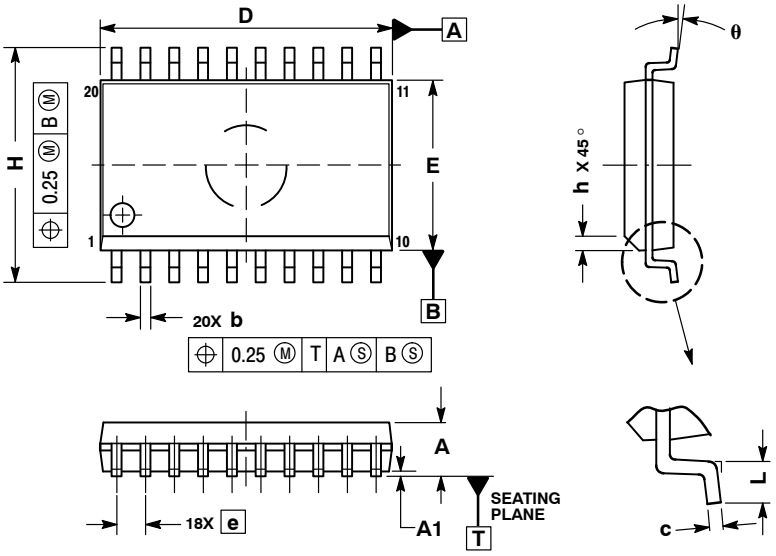
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SOIC−20 WB
CASE 751D−05
ISSUE H

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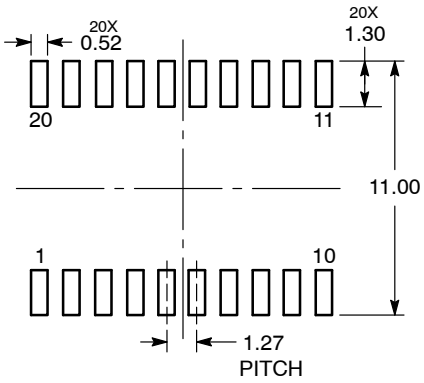
SCALE 1:1



NOTES:
1. DIMENSIONS ARE IN MILLIMETERS.
2. INTERPRET DIMENSIONS AND TOLERANCES PER ASME Y14.5M, 1994.
3. DIMENSIONS D AND E DO NOT INCLUDE MOLD PROTRUSION.
4. MAXIMUM MOLD PROTRUSION 0.15 PER SIDE.
5. DIMENSION B DOES NOT INCLUDE DAMBAR PROTRUSION. ALLOWABLE PROTRUSION SHALL BE 0.13 TOTAL IN EXCESS OF B DIMENSION AT MAXIMUM MATERIAL CONDITION.

	MILLIMETERS	
DIM	MIN	MAX
A	2.35	2.65
A1	0.10	0.25
b	0.35	0.49
c	0.23	0.32
D	12.65	12.95
E	7.40	7.60
e	1.27 BSC	
H	10.05	10.55
h	0.25	0.75
L	0.50	0.90
θ	0°	7°

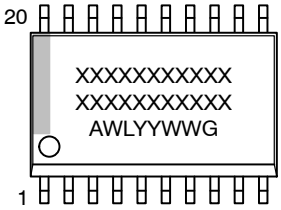
RECOMMENDED SOLDERING FOOTPRINT*



DIMENSIONS: MILLIMETERS

*For additional information on our Pb−Free strategy and soldering details, please download the **onsemi** Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

GENERIC MARKING DIAGRAM*



XXXXX = Specific Device Code
A = Assembly Location
WL = Wafer Lot
YY = Year
WW = Work Week
G = Pb−Free Package

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “▪”, may or may not be present. Some products may not follow the Generic Marking.

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